

Android Project : OTAMS

SEG 2105 A00 - Introduction To Software Engineering Fall 2025 University of Ottawa

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Introduction :

The Online Tutoring Appointment Management System (OTAMS) is an Android application developed to efficiently improve the scheduling process at the University of Ottawa. The application serves three user roles : student, tutor and administrator. Each role has a respective use, for example, the student is looking for tutoring services, the tutor provides the tutoring services and the administrator manages the user registrations. This application facilitates the entire lifecycle of tutoring sessions, from account creation and availability management to booking, approving and rating systems. Moreover, it integrates user authentication, sessions management, availability tracking and a detailed approval workflow, all supported by a database backend.

Updated UML class diagram :

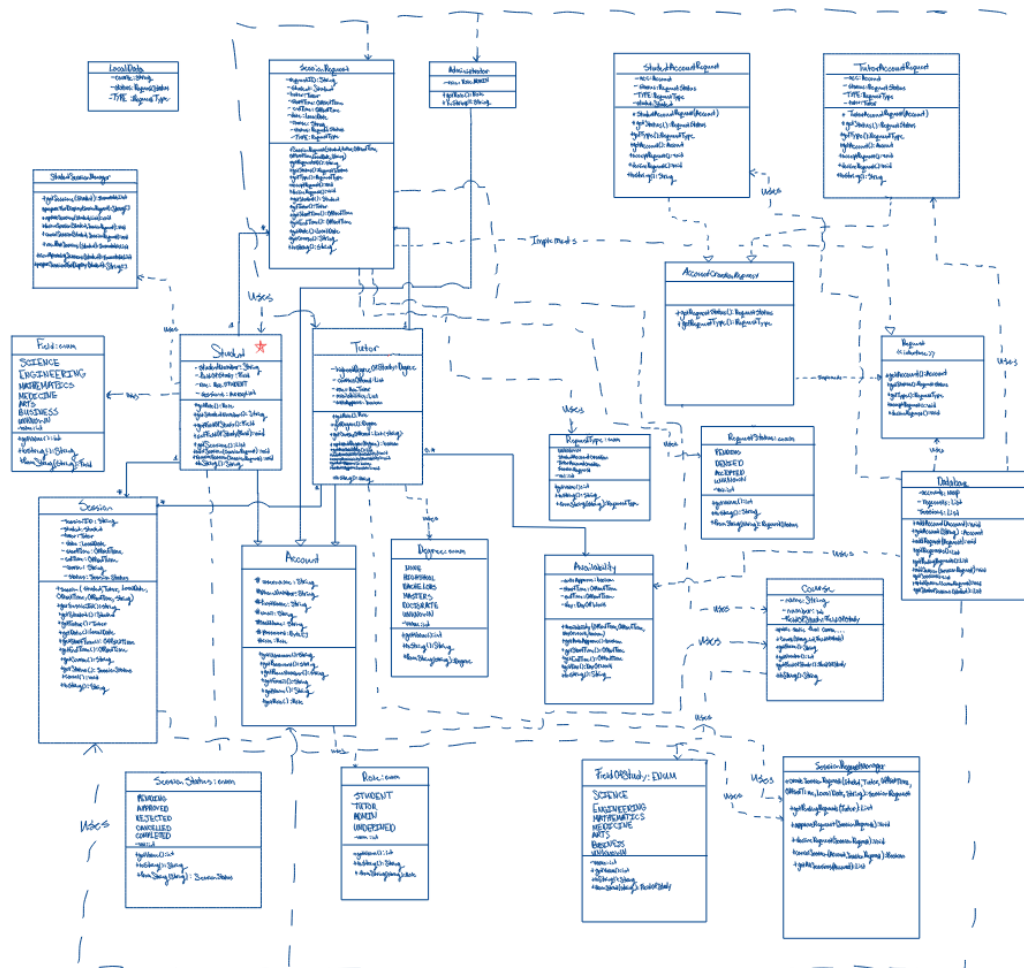


Fig 1. Updated UML Diagram, for a better version, please check the Deliverable 4 folder on the GitHub repository.

Team member contribution :

Task :	Member Name :
Deliverable 1	
- Created student registration UI layout	Daniil
- Implemented input fields	Anastasia
- Added validation for student fields	
- Implemented email validation	
- Created student class	Karim, Sébastien
- Created tutor registration UI layout	Daniil
- Implement input fields	Anastasia
- Created welcome page UI	Meagan
- Implement Tutor class	Karim, Sébastien
- Set-up database	Vincent, Sébastien
- Created login UI	Meagan
- Implemented login functionality for the roles	Sébastien
- Configured the .gitignore	Sébastien
- Wrote README	Sébastien
- Created demo video	Sébastien

- Created Administrator account	Sébastien
Deliverable 2	
- Pending requests	Sébastien
- Implemented registration request class	Sébastien
- Create database for storing requests	Sébastien
- Updated student and tutor registration to submit requests	Sébastien, Daniil
- Added request status enum	Sébastien
- Created administrator welcome screen	Sébastien
- Implemented status page UI	Meagan
- Implemented registration request UI	Anastasia
- Created recycler view for displaying requests	Sébastien
- Display all user info except password	Sébastien
- Added approve and reject buttons	Anastasia
- Implemented approve request functionality	Sébastien
- Created rejected requests view	Anastasia
- Updated database for status changes	Sébastien
- Changed login to check approval status	Sébastien

- Implemented pending approval message	Sébastien
- Implemented rejection message with phone number	Anastasia
- Updated welcome screen for approved users	Sébastien
- Added homepage buttons and logic	Meagan
- Sending email to user upon approval/rejection (NOT USED)	Vincent
- Updated UML diagram	Anastasia, Sébastien
- Created demo video	Sébastien
Deliverable 3	
- Create availability for the time slots	Karim, Vincent, Sébastien
- Designed create availability slot UI and logic	Meagan
- Added past date validation	Sébastien, Daniil
- Create database methods for storing slots	Sébastien
- Added error messages for validation errors	Daniil
- Implemented view upcoming tutor sessions UI and logic	Meagan, Daniil
- Implemented date and time formatting	Sébastien, Daniil
- Created tutor calendar UI and Logic	Daniil

- Implemented upcoming sessions	Sébastien
- Implemented past sessions	Sébastien
- Added data storing functionality	Sébastien
- Created pending session requests	Vincent, Karim, Daniil
- Added cancel session functionality	Sébastien
- Updated status session in database	Sébastien
- Updated UML diagram	Sébastien, Meagan
- Updated README	Sébastien
- Created demo video	Sébastien
Deliverable 4	
- Created student dashboard UI and logic	Meagan, Daniil
- Implemented search by course code	Sébastien, Vincent
- Implemented available slots display	Sébastien, Daniil
- Implemented upcoming sessions for students	Sébastien, Daniil, Vincent
- Implemented students requesting sessions	Sébastien, Daniil
- Implemented rating tutor functionality	Sébastien
- Implemented rating tutor UI	Didn't get done
- Implemented students viewing the status of their request	Sébastien, Daniil

- Recorded demo video	Sébastien
- UML Diagram	Sébastien, Meagan
- Implemented Unit Tests	Vincent
- Took screenshots of the app	Sébastien

Lessons learned :

Meagan:

As someone who had never used Android Studio before, this project provided valuable experience with a new IDE. Initially I found creating XML layouts tedious and unintuitive. However, by the second deliverable, I found it became more intuitive as I got used to the relationships between XML attributes and UI components. Working with a team presented both challenges and more opportunities to learn. Having only collaborated with a partner once before, coordinating with multiple team members required developing better communication skills. Nonetheless, it was interesting to see how differently people worked, it certainly made the project a bit more challenging. On top of learning how to use a new IDE, I got a chance to touch on UI and the logic behind the UI as well as UML diagrams, and I am very glad I did so. I'm positive that this experience will come in handy in the future. Overall, I have learned a lot thanks to this project and I'm very glad to have worked with a team that was adaptable and accommodating.

Karim:

During this semester, I got the opportunity to implement the fundamentals I learned from the SEG 2105 and ITI 1121. Though, I'm still a long deal away from becoming optimal in a team setting as I had issues with communication where sometimes, instead of doing what's expected of me only, I went ahead and thought I did a good job by doing more, but sometimes, I don't even do the bare minimum because I got overwhelmed sometimes, for example, during deliverable 2, I failed to do any meaningful work because it was during the midterms. This project definitely taught me important lessons about working in a team setting, something that was way outside my comfort zone. Could I have done better? Definitely. I could've started off with the same transparency I showed in the later deliverables, like deliverable 3 and 4, where I worked on the code for the Availability class, Session Request, and that would've shown what I'm really capable of. Overall, the shortcomings in this project definitely humbled and helped me become a better engineering student

Vincent:

I have gained valuable insight into the field of software engineering during the entire semester. From exploring database implementation to understanding Android code interactions, these experiences have provided me with a comprehensive overview of the discipline. It has pushed my understanding of coding further than ever before. However, I believe the biggest lesson I learned was that software engineering does not align with my interests or career aspirations, and I do not want to work in this field at all.

Anastasia

While working on this project, I have learned many things that will come to be useful in the future. I have learned how complex it is working in a team, making sure everyone is on the same page and does their part. I also learned how important communication is when it comes to such projects. Since communication helps avoid any misunderstandings and potential wasted time, I found it essential to

maintain a constant line of communication. Regarding material I have learnt, I acquired the knowledge of how to use new softwares. These include Github and Android Studio. Overall, this project proved to be valuable in many aspects, both in a positive and negative manner.

Sébastien:

This project was a very interesting challenge, though simple at first glance it became clear that it wasn't quite as simply as originally expected, however, the real challenge would come with learning to program with other people who have different mental images of what the end goal is. This was both a blessing and a curse, causing both great joy in learning about how others do things and what's the best solution, but also great misery when someone pours their heart and soul into something only to find that it wasn't quite what we were aiming for as a group. We did eventually figure it all out through mock designs and conversations about how the app should flow. Ultimately this is just the experience of working as a team, and it was quite the up and down, but overall enjoyable. I just hope to never have to touch UI code again...

Daniil:

At first glance, this project seemed quite simple and straightforward, but not for long. As the project started, I got to discover what Android Studio has to offer and learn how XML layout and corresponding activity/logic come together to make the front-end of most Android apps. However, as time went on, I understood that even simple things like a calendar layout and logic can become increasingly complex as more features are added, which turned chill coding evenings into somewhat exhausting all-nighters. By this point, I learned that communication is key, as it would have saved me from doing so much extra work (learned the hard way). Closer to the end, I discovered that the assigned roles(UI, Database, etc) become a formality once your features have to be integrated with the work of others. I quickly went from working on UI layouts and logic to learning how our Firebase is set up to implement my features. By the last deliverable, I understood that the most complex part of this project is understanding the code/logic of other people and what they had envisioned, as changing one little thing in the logic of others to fix/implement something in yours echoed throughout the entire project, breaking other things. This cycle of breaking and fixing kinda showed what

software development is all about, as by the end, I had a mental map of how all things were connected within the project engraved in my mind, which actually resulted in improvement of communication with other team members, as now I understood what struggles they faced throughout their work. However, I hope that I will never have to make a calendar from the ground up ever again... I still have nightmares about it.

App screenshots :

OTAMS

Login

Sign-up

Figure 2 : *Login page*

Register as...

Tutor

Student



Figure 3 : *Registration page*

Student Form

Username

Password

Phone number

Email

First Name

Last Name

Apply



Figure 4 : *Student registration forum*

Tutor Form

Username

Password

Phone number

Email

First Name

Last Name

Science

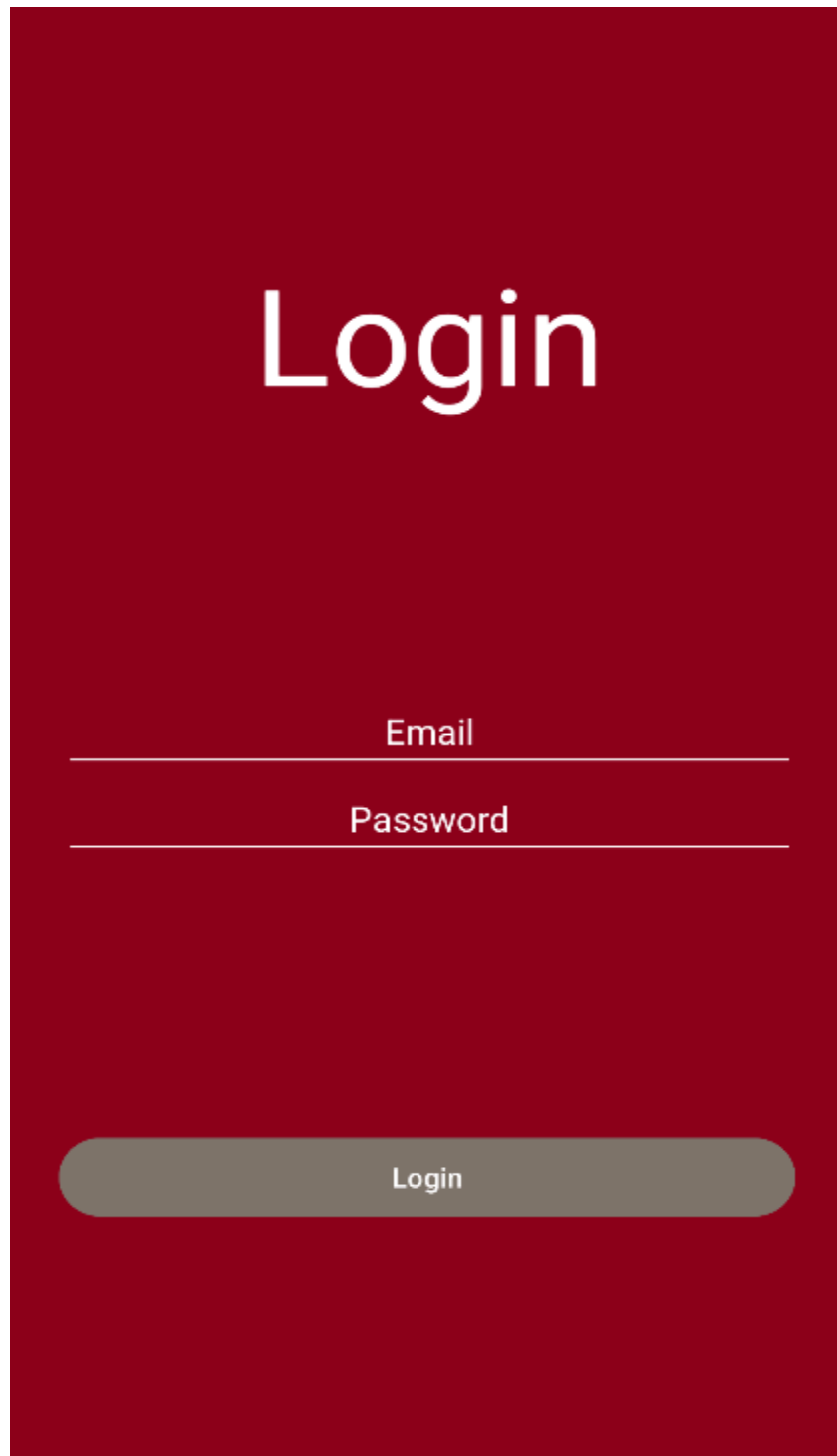


Apply

NONE



Figure 5 : *Tutor registration forum*

The image shows a login page with a solid dark red background. At the top center, the word "Login" is written in a large, white, sans-serif font. Below this, there are two input fields. The first field is labeled "Email" in a small, white, sans-serif font, with a white horizontal line underneath it. The second field is labeled "Password" in a small, white, sans-serif font, also with a white horizontal line underneath it. At the bottom center, there is a dark gray, rounded rectangular button with the word "Login" written in a small, white, sans-serif font.

Login

Email

Password

Login

Figure 6 : *Login page*

You are logged
in as God I hate
android studio.
Your application
for Tutor is still
under review,



Figure 7 : *Application under review page*

Pending Requests

Henry Tan
Username: H
Role requested: Student
God I hate android studio
Username: ornana
Role requested: Tutor

[Go to rejected requests](#)



Figure 8 : *Admin pending requests page*

12:47

You are currently logged in as admin

Information Page

First name: God I hate android

Last name: studio

Username: ornana

Email: akfwajkfma@gmail.com

Doctor of Science

Accept Request

Decline Request



Go to pending

Go to rejected

Figure 9 : *Admin application review page*

Rejected Requests

Henry Tan
Username: Henry
Role requested: Student
God I hate android studio
Username: ornana
Role requested: Tutor

Go to pending requests



Figure 10 : *Admin denied requests page*

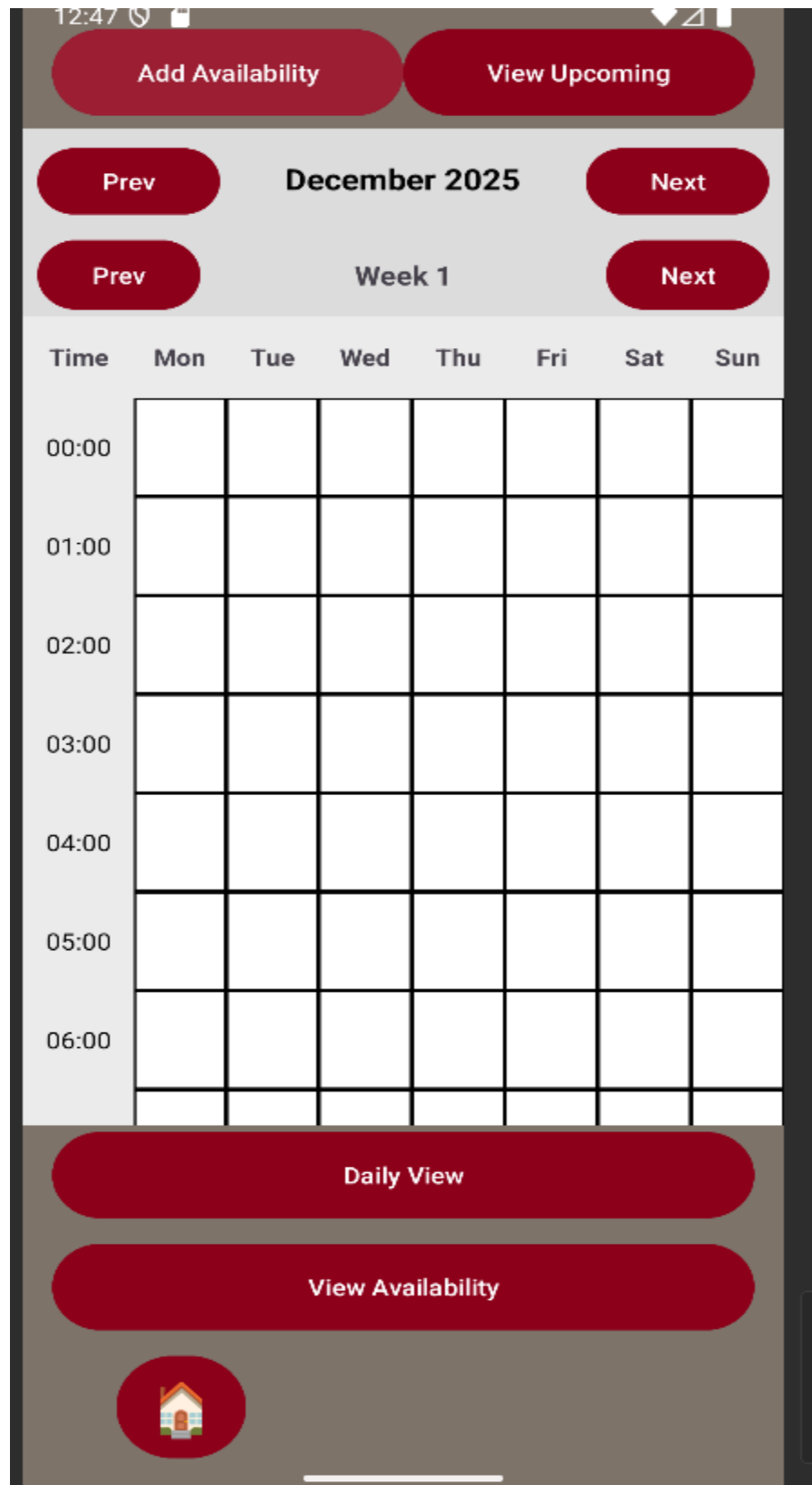


Figure 11 : *Tutor weekly calendar view page*

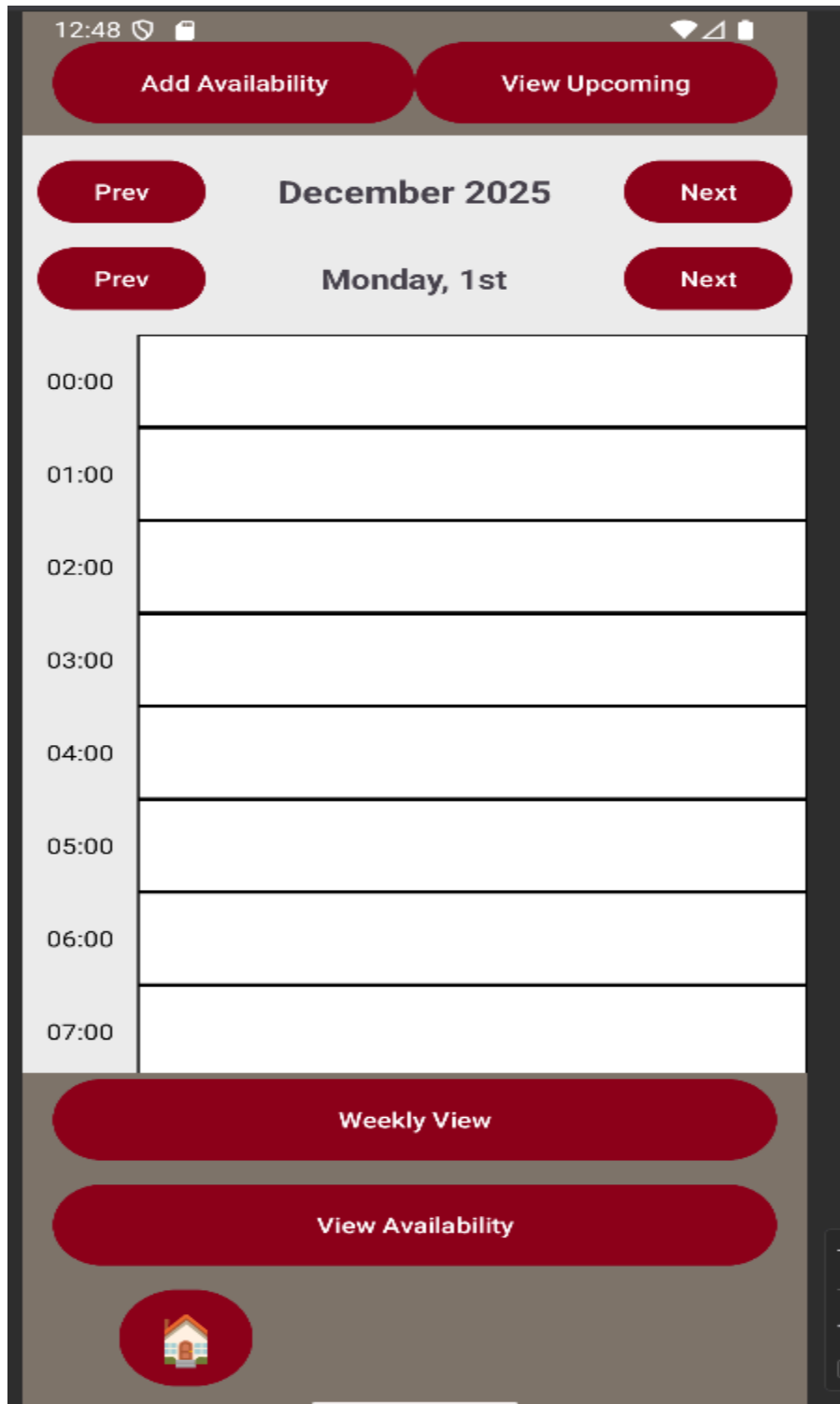


Figure 12 : *Tutor daily calendar view page*

12:48

MONDAY

Start Time :

End Time :

☐ Automatically approve sessions

Add availability

View calendar

Figure 13 : *Tutor add availability page*

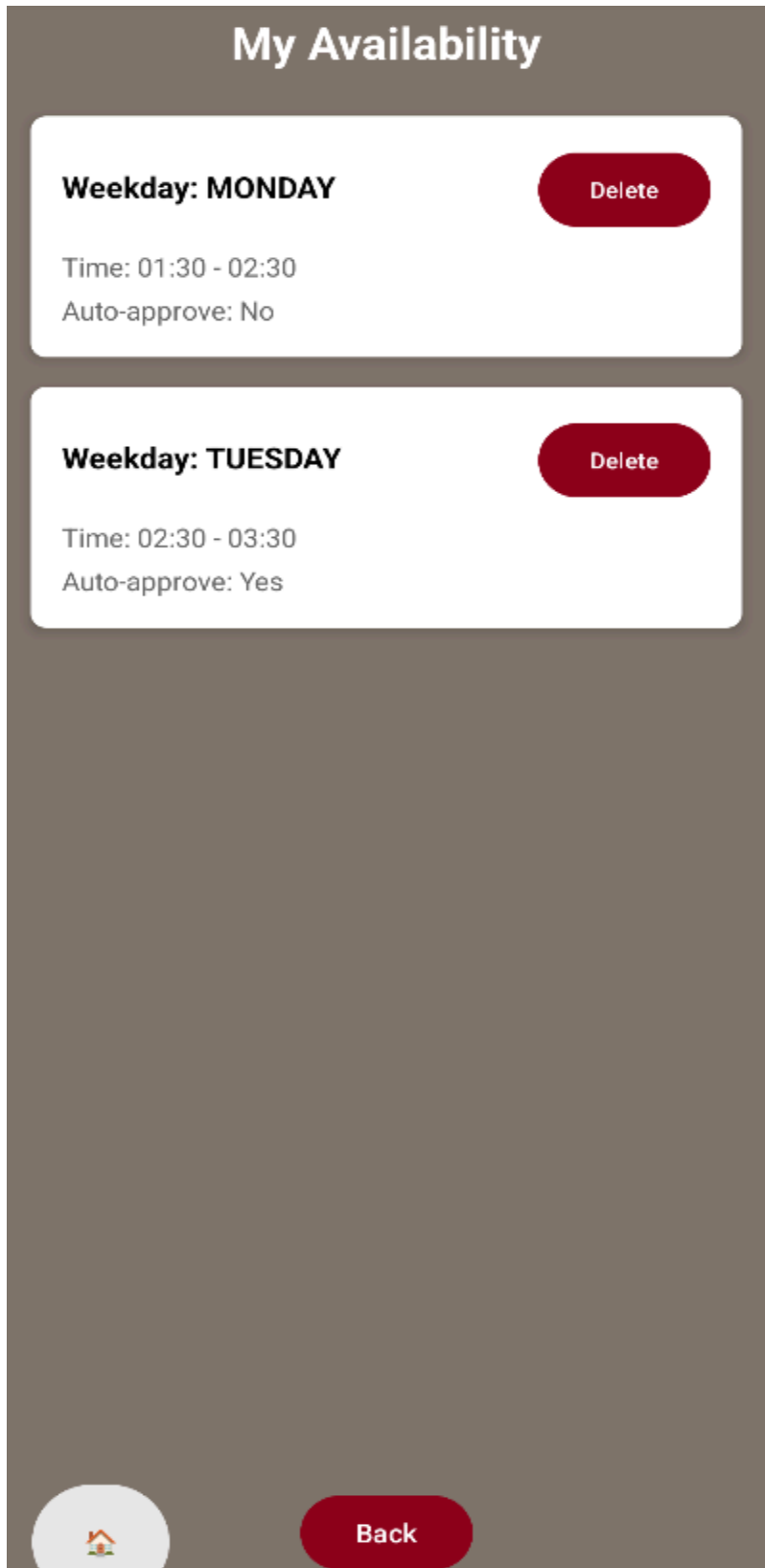


Figure 14 : *Tutor view availability page*

Your Upcoming Sessions

Session with:
studentTest
On 2025-12-08T00:00-05:00

Back

View
Pending

Past
Sessions



Figure 15 : *Tutor upcoming sessions page*

Pending Requests

Session with:
studentTest
On 2025-12-08T00:00-05:00



Figure 16 : *Tutor pending requests page*

Information Page

Session with:

studentTest

On: 2025-12-08T00:00-05:00

Starts at: 01:30-05:00

End at: 02:30-05:00

Accept Request

Decline Request



Go to pending

Go to rejected

Figure 17 : *Tutor session information page*



Figure 19 : *Tutor view past sessions page*

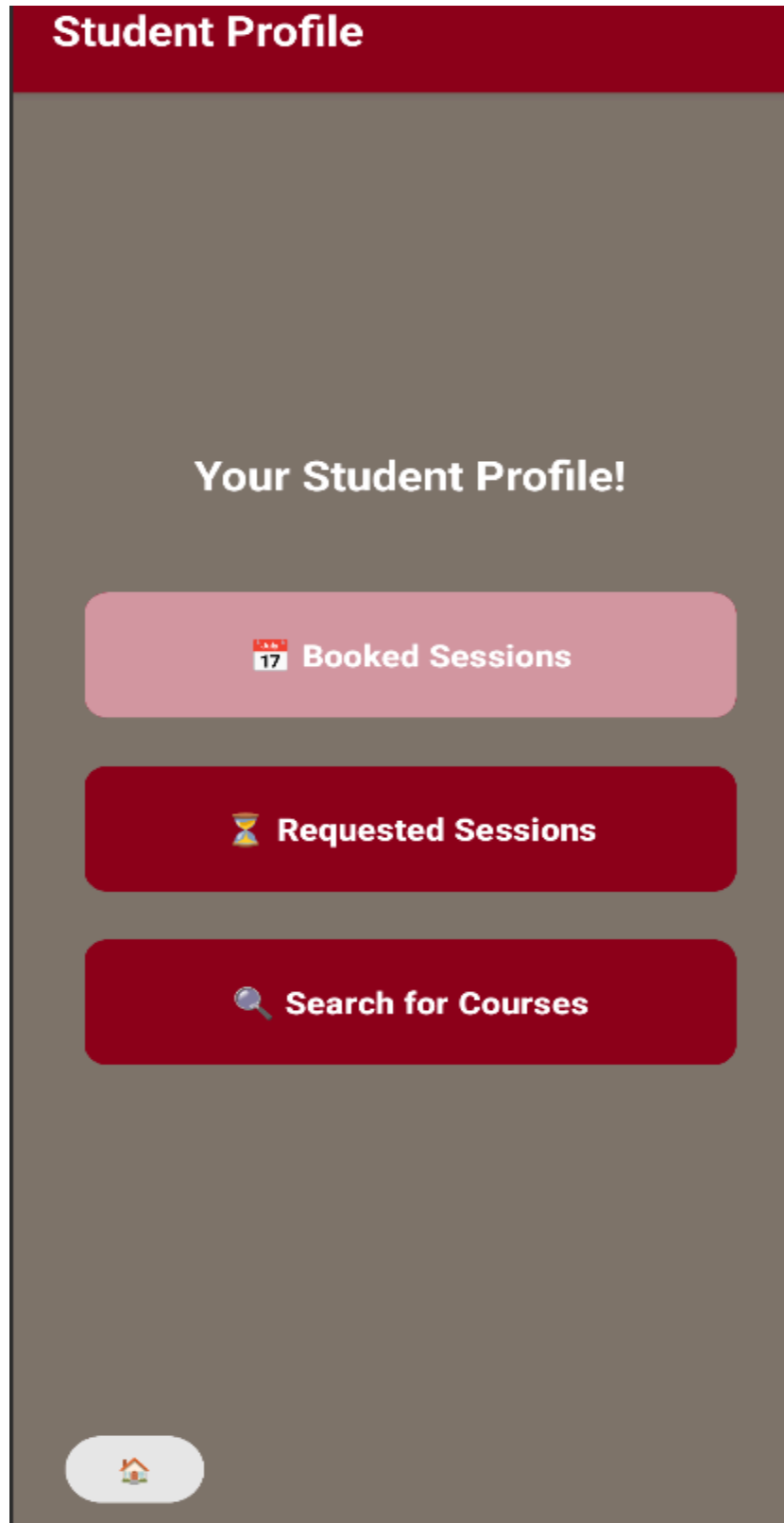


Figure 20 : *Student welcome page*

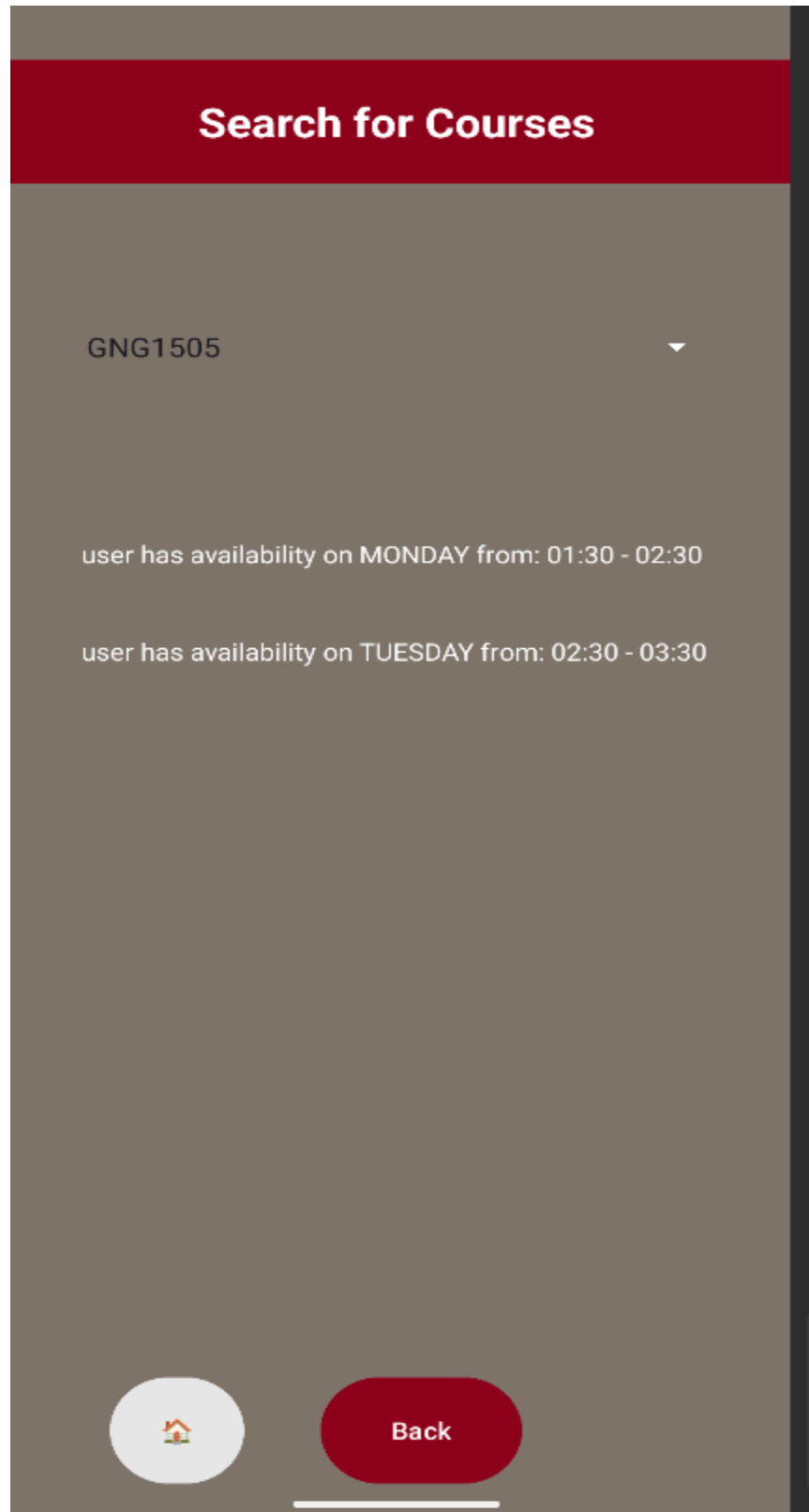


Figure 21 : *Student search for courses page*

Booking Session

user has availability on MONDAY
from: 01:30 - 02:30

Start Time

End Time

Request



Back

Figure 22 : *Student request session page*



Figure 23 : *Student booked sessions page*

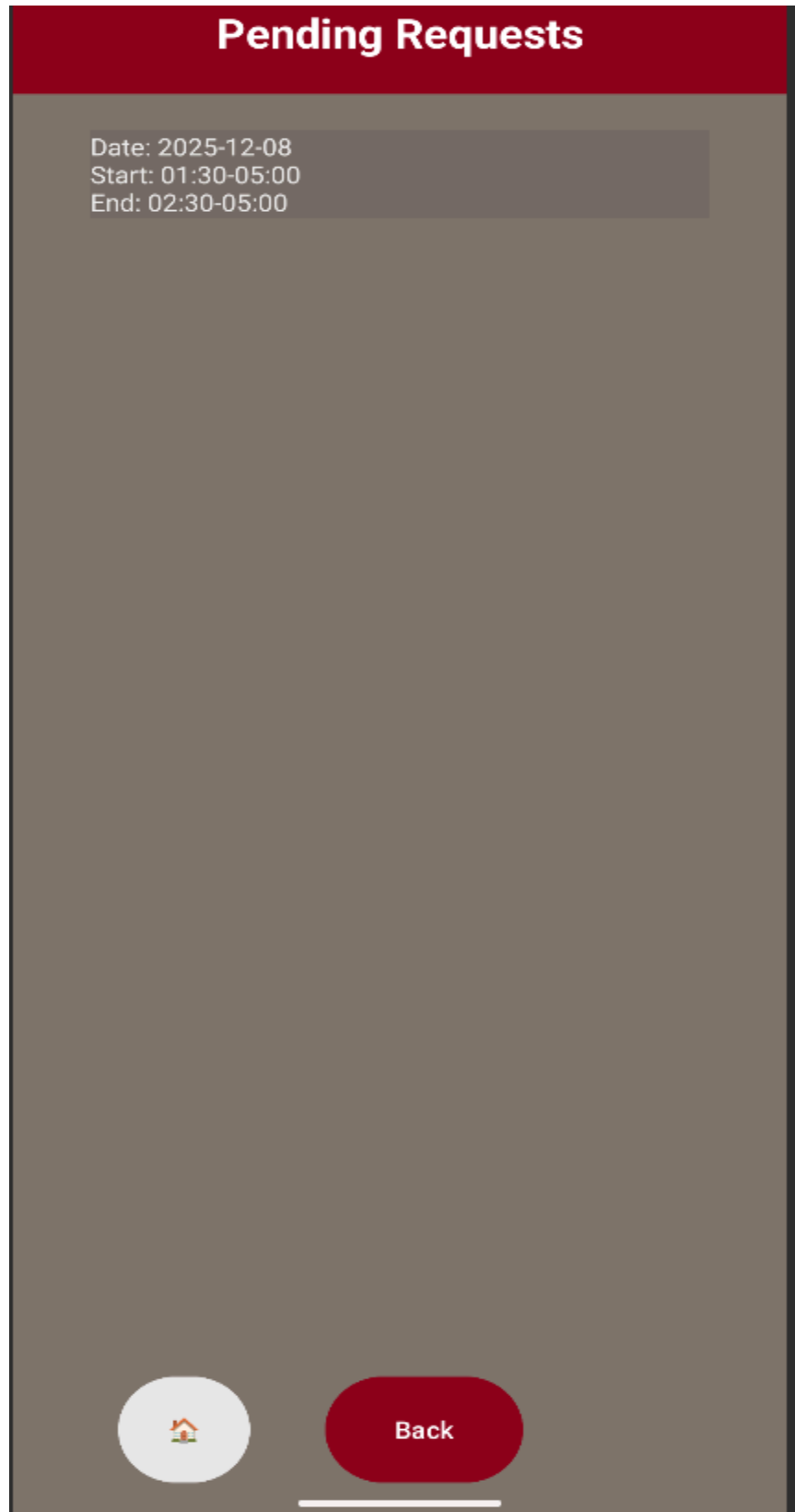


Figure 24 : *Student pending sessions page*